

QUESTION 1

$$11001111_2 + 1101001_2 =$$

- A. 470_8 B. 468_8 C. 367_8 D. 312_8 E. None of these

QUESTION 2

Which of the following data types can *not* be assigned the value 0?

- A. `bool` B. `int` C. `char *` D. `double` E. None of these

QUESTION 3

If `x` has the value 5, what is the value of this expression?

$$(x=4)?1:-1$$

- A. 5 B. 4 C. 1 D. -1 E. None of these

QUESTION 4

What is the value of this expression?

$$85/7*7+85\%7$$

- A. 7 B. 86 C. 85 D. 0 E. None of these

QUESTION 5

Which of the following expressions using a function from `math.h` gives an `int` which is the absolute value of `int i`?

- I. `sqrt(i^2)` II. `abs(i)` III. `sqrt(i*i)`

- A. I only B. II only C. III only D. II and III E.

QUESTION 6

Suppose algorithms A and B solve the same problem. If algorithm A runs in time proportional to the number of inputs, and algorithm B runs in time proportional to the square of the number of inputs, what can we say about the running times of algorithms A and B?

- A. On all computers and all inputs, algorithm A is faster
B. On all computers, for sufficiently small inputs, algorithm B is faster
C. On most computers and all inputs, algorithm B is faster
D. On all computers, for sufficiently large inputs, algorithm A is faster
E. More than one of these

For the following questions, give the output of the code to the right for the given input.

QUESTION 7

5 0 0
0 5 0
0 0 5
0 0 5

- A. Winner is A
- B. Winner is B
- C. Winner is C
- D. Winner is a
- E. None of these

QUESTION 8

3 0 2
5 0 0
0 2 3
0 5 0

- A. Winner is A
- B. Winner is B
- C. Winner is C
- D. Winner is a
- E. None of these

QUESTION 9

0 6 0 -1 4 2 1 3 1
2 3 0 2 0 3 1 0 4
1 1 1 2 2 2 1 3 1

- A. Winner is A
- B. Winner is B
- C. Winner is C
- D. Winner is a
- E. None of these

```
int [3];
int total[3];

int a, b, c;

while (cin >> a >> b >> c) {
    if (a<0 || b<0 || c<0 ||
        a+b+c != 5) continue;
    if (a>0) votes[0]++;
    if (b>0) votes[1]++;
    if (c>0) votes[2]++;
    total[0]+=a;
    total[1]+=b;
    total[2]+=c;
}

char w = 'Z';

if (votes[0]>votes[1] &&
    votes[0]>votes[2]) w='A';
else if (votes[1]>votes[0] &&
    votes[1]>votes[2]) w='B';
else if (votes[2]>votes[0] &&
    votes[2]>votes[1]) w='C';
else if (total[0]>total[1] &&
    total[0]>total[2]) w='a';
else if (total[1]>total[0] &&
    total[1]>total[2]) w='b';
else if (total[2]>total[0] &&
    total[2]>total[1]) w='c';

cout << "Winner is " << w << endl;
```

QUESTION 10

Which of the following declares `r` to be of type `random`?

- A. `new random r();`
- B. `random r;`
- C. `r = random();`
- D. `random r();`
- E. None of these

QUESTION 11

What kind of data member is `seed`?

- A. `public`
- B. `protected`
- C. `private`
- D. `default`
- E. None of these

QUESTION 12

Suppose you were writing the function `nextInt()`. Which of the following returns all possible values in the correct range?

- A. `return max*next;`
- B. `return max*next();`
- C. `return int(max+next());`
- D. `return int(max*next() + 1);`
- E. None of these

QUESTION 13

What should replace `<*1>` in the code to the right?

- A. `bool`
- B. `int`
- C. `float`
- D. `double`
- E. None of these

QUESTION 14

Assuming that `<*1>` is filled in correctly, which of these would return `true`?

- A. `is_palindrome("abba")`
- B. `is_palindrome("murmur")`
- C. `is_palindrome("Candelabra")`
- D. `is_palindrome("Eve")`
- E. None of these

`random.h:`

```
class random {
```

```
    long seed;
```

```
public:
```

```
    random();
```

```
    random(long s);
```

```
    // next() returns a random
```

```
    // number r with 0<=r<1
```

```
    double next();
```

```
    // nextInt() returns a number
```

```
    // between 1 and max
```

```
    int nextInt(int max);
```

```
}
```

```
<*1> is_palindrome(apstring s) {
    int l = s.length();
    for (int i=0; i<l/2; ++i)
        if (s[i]!=s[l-i])
            return false;
    return true;
}
```

QUESTION 15

What is returned by the function call `mystery(6,0)`?

- A. 36 B. 6 C. 360
D. 21 E. None of these

```
int mystery(int x, int y) {
    if (x==0) return y;
    else return mystery(x-1,y+x);
}
```

QUESTION 16

What is output by the code to the right on this input?

This is a test.
This is only a test.

- A. This
B. This is a test.
C. This is a test. (followed by a new line character)
D. This is a test.
This is only a test. (followed by a new line character)
E. None of these

```
apstring s;
getline(cin, s);
cout << s;
```

QUESTION 17

Which of these could replace `<*1>` as a syntactically correct call to the function `g`?

- A. `g(&y,&x)` B. `g(3,7)`
C. `g(x,y)` D. `g(*x,*y)`
E. None of these

```
void g(int * x, int *y) {
    // some code
}

int main() {
    int x, int y;
    <*1>
}
```

QUESTION 18

Which of the following correctly fills `<*1>` in the code to the right so that if `ch` is 'a' or 'A', it is processed by the first case, and so on?

- A. `islower(ch)` B. `isupper(ch)`
C. `tolower(ch)` D. `toupper(ch)`
E. None of these

```
char ch;
cin >> ch;

switch(<*1>) {
case 'A': //first case
    //some code
    break;
case 'B': //second case
    //some code
    break;
...
}
```

QUESTION 19

Which of these replaces **<*1>** in the code to the right to declare PI to have the value 4 times the arctangent of 1.0, and that the value of PI cannot be changed?

- A. `const double PI = 4*atan(1.0);`
- B. `#define PI = 3.14`
- C. `double PI = 3.14`
- D. `const double PI = tan(4.0);`
- E. None of these

For the remaining questions, assume **<*1>** is filled in correctly.

QUESTION 20

If s is a shape with 4 sides that have length 2, what is the return value of `s.area()`?

- A. 2
- B. 4
- C. 8
- D. 12.56
- E. None of these

QUESTION 21

What is the running time for a call to `area()`? Choose the smallest correct answer

- A. $O(s)$
- B. $O(s+l)$
- C. $O(1)$
- D. $O(s*l)$
- E. None of these

QUESTION 22

Which of the following will sum entries read from input, stopping at the first negative number? The negative number should not be added to the sum. Assume in all cases that x and sum are previously declared integer variables with sum initialized to 0.

- A.

```
cin >> x;
while(x>=0) {
    sum+=x;
    cin >>x;
}
```
- B.

```
for(cin>>x;x>=0;cin>>x)
    sum+=x;
```
- C.

```
do {
    cin >> x;
    sum+=x;
} while (x>=0);
```
- D. Both A and B
- E. A, B, and C

shape.h:

```
class shape {
public:
    shape(int sides, int length);
    int perimeter() const;
    double area() const;
private:
    int s;
    int l;
};
```

shape.cpp:

```
#include "shape.h"
#include <math.h>
<*1>

shape::shape(int sides,
             int length) const {
    s=sides;
    l=length;
}

int shape::perimeter() const {
    return s*l;
}

double shape::area() {
    double a = l*tan(PI/2-PI/s)*l/2;
    return 0.5*a*perimeter();
}
```

QUESTION 23

Which of these sets book b to have title "Anatomy" and author "Gray"?

- A. `b("Anatomy", "Gray");`
- B. `b(title=Anatomy, author=Gray);`
- C. `b.title="Anatomy"; b.author="Gray";`
- D. `b->title=Anatomy; b->author=Gray;`
- E. None of these

```
struct book {
    apstring title;
    apstring author;
}
```

QUESTION 24

Which of these declares a variable library which can hold ten books?

- A. `apvector<book> library(10);`
- B. `apvector library(book, 10);`
- C. `apvector library(10, book);`
- D. `apvector<10> library(book);`
- E. A, B, and C

QUESTION 25

What is output by the code to the right?

- A. 4
- B. 3
- C. 2
- D. Nothing; the code has a syntax error
- E. None of the above

```
int x=1;

void f(int &x) { ++x; }

int main() {
    int x=2;
    {
        int x=3;
        f(x);
    }
    cout << x;
}
```

QUESTION 26

What happens when a constructor is not defined for a user-defined class?

- A. You cannot instantiate the class.
- B. There is a default constructor, which takes arguments of the same type as the data members in order.
- C. There is a default constructor which initializes data members with basic types to 0 and with class types by calling their no argument constructor.
- D. There is a default constructor which does not initialize the data members.
- E. None of these.

QUESTION 27

Which of these could replace `v[i]/=2` without changing the behavior of `half()` for vectors of positive numbers.

- A. `v[i]*=1/2` B. `v[i]--` C. `v[i]*=0.5`
 D. Both A and C E. None of these

```
void half(apvector<int> v) {
    for (int i=0; i<v.length(); ++i)
        v[i]/=2;
}
```

QUESTION 28

What is output by the code to the right?

- A. 2.53.79.9 B. 2.5
 C. 3.7 D. 9.9
 E. None of these

```
apstack<double> s;
double d;
s.push(2.5);
s.push(3.7);
s.push(9.9);
d = s.top();
cout << d;
```

QUESTION 29

Suppose that `apmatrix<int> m` is initialized as below. What is the value of the expression `m[1]`?

1	2	3	4
5	6	7	8
9	10	11	12

- A.

1

 B.

1	2	3	4
---	---	---	---

 C.

6

 D.

1
5
9

 E. None of these

QUESTION 30

Which of these sorts has a partition step which places all items smaller than the pivot before the pivot and all items larger than the pivot after the pivot?

- A. Insertion B. Selection C. MergeSort D. QuickSort E. None of these

QUESTION 31

Which of these is a correct call to search attempting to locate an element with the value 5 in an array A with length 14?

- A. A.search(5,14)
- B. A.search(14,5)
- C. search(A,5,14)
- D. search(A,14,5)
- E. None of these

```
int search(int v[], int l, int z)
{
    for(int i=0; i<l; ++i)
        if(v[i]==z) break;
    if (i==l) i=-1;
    return i;
}
```

QUESTION 32

In the search above, if A[7] is 5, and 5 does not appear anywhere else in A, how many times will an element of A be compared with the value 5 before returning an answer?

- A. 0
- B. 14
- C. 7
- D. 8
- E. None of these

QUESTION 33

What replaces **<*1>** in the code to the right to declare a variable named in which is associated with the input file whose name is given in name?

- A. ifstream in(name);
- B. ofstream in(name);
- C. ifstream in = new ifstream(name);
- D. ofstream in = new ofstream(name);
- E. None of these

```
void negate(apstring name) {
    <*1>
    int x;
    while(!in.eof()) {
        in>>x;
        cout << -x;
    }
}
```

QUESTION 34

What will be the output of the function call negate("test.in") when the file test.in contains the data below?

```
1 2
3 4 5
16
```

- A. -1-2-3-4-5-16
- B. -1 -2 -3 -4 -5 -16
- C. -1-2
- D. -1 -2
- E. None of these

QUESTION 35

Which of these expressions is represented by the truth table below?

a	b	c
0	0	0
0	1	0
1	0	0
1	1	0

- A. $a || !b$ B. $a \&\&!a$ C. $a \&\&!b$
 D. $!a || b$ E. None of these

QUESTION 36

For the function to the right to implement the Merge Sort algorithm, what must the function `Merge()` do?

- A. Nothing; it shouldn't be there
 B. Sort A
 C. Combine the unsorted list stored in the front half of A with the unsorted list in the back half of A
 D. Combine the sorted list stored in the front half of A with the sorted list in the back half of A
 E. None of these

```
void MergeSort(int A[], int front,
               int back)
{
    int mid=(front+back)/2;
    if (mid==front) return;
    MergeSort(A,front,mid);
    MergeSort(A,mid,back);
    Merge(A,front,back);
}
```

QUESTION 37

Assume that `Merge()` is correctly and efficiently implented. What is the worst case running time of `MergeSort()`, where n is the length of array A? Choose the smallest correct answer.

- A. $O(\log n)$ B. $O(n)$ C. $O(n \log n)$
 D. $O(n^2)$ E. None of these

QUESTION 38

Which of these expressions has the value 3?

- A. `A[3]` B. `&(A[3])` C. `1+&(A[3])`
 D. `*A+3` E. None of these

```
int A[] = {5, 4, 3, 2, 1};
```

QUESTION 39

What is returned by $f(-5)$?

- A. -5 B. 0 C. 5
D. 16 E. None of these

```
int f(int x) {  
    if (x<=0) return 0;  
    else return g(x-1,x-1);  
}  
  
int g(int y, int z) {  
    if (y==0) return z;  
    else return g(y-1,f(z));  
}
```

QUESTION 40

What is returned by $f(3)$?

- A. -5 B. 0 C. 5
D. 16 E. None of these

AP QUICK REFERENCE

NOTE: The front and back of this page includes reference information for the AP classes. This is the same reference information that students use during the AP exam. You may use this reference sheet during the contest. You may detach this page from the test booklet.

Quick Reference for apstring

extern const int npos; // used to indicate not a position in the string

class apstring

// public member functions

// constructors/destructor

```
apstring(); // construct empty string ""
apstring(const char * s); // construct from string literal
apstring(const apstring & str); // copy constructor
~apstring(); // destructor
```

// assignment

```
const apstring & operator= (const apstring & str); // assign str
const apstring & operator= (const char * s); // assign s
const apstring & operator= (char ch); // assign ch
```

// accessors

```
int length() const; // number of chars
int find(const apstring & str) const; // index of first occurrence of str
int find(char ch) const; // index of first occurrence of ch
apstring substr(int pos, int len) const; // substring of len chars, starting at pos
const char * c_str() const; // explicit conversion to char *
```

// indexing

```
char operator[ ](int k) const; // range-checked indexing
char & operator[ ](int k); // range-checked indexing
```

// modifiers

```
const apstring & operator+= (const apstring & str); // append str
const apstring & operator+= (char ch); // append char
```

// The following free (non-member) functions operate on strings

// I/O functions

```
ostream & operator<< ( ostream & os, const apstring & str );
istream & operator>> ( istream & is, apstring & str );
istream & getline( istream & is, apstring & str );
```

// comparison operators

```
bool operator== ( const apstring & lhs, const apstring & rhs );
bool operator!= ( const apstring & lhs, const apstring & rhs );
bool operator< ( const apstring & lhs, const apstring & rhs );
bool operator<= ( const apstring & lhs, const apstring & rhs );
bool operator> ( const apstring & lhs, const apstring & rhs );
bool operator>= ( const apstring & lhs, const apstring & rhs );
```

// concatenation operator +

```
apstring operator+ ( const apstring & lhs, const apstring & rhs );
apstring operator+ ( char ch, const apstring & str );
apstring operator+ ( const apstring & str, char ch );
```

AP QUICK REFERENCE CONT.

Quick Reference for apvector and apmatrix

```
template <class itemType>
class apvector

// public member functions

// constructors/destructor
apvector(); // default constructor (size==0)
apvector(int size); // initial size of vector is size
apvector(int size, const itemType & fillValue); // all entries == fillValue
apvector(const apvector & vec); // copy constructor
~apvector(); // destructor

// assignment
const apvector & operator= (const apvector & vec);

// accessors
int length() const; // capacity of vector

// indexing
itemType & operator[ ](int index); // indexing with range checking
const itemType & operator[ ](int index) const; // indexing with range checking

// modifiers
void resize(int newSize); // change size dynamically; can result in losing values
```

```
template <class itemType>
class apmatrix

// public member functions

// constructors/destructor
apmatrix(); // default size is 0 x 0
apmatrix(int rows, int cols); // size is rows x cols
apmatrix(int rows, int cols, const itemType & fillValue); // all entries==fillValue
apmatrix(const apmatrix & mat); // copy constructor
~apmatrix( ); // destructor

// assignment
const apmatrix & operator = (const apmatrix & rhs);

// accessors
int numrows() const; // number of rows
int numcols() const; // number of columns

// indexing
const apvector<itemType> & operator[ ](int k) const; // range-checked indexing
apvector<itemType> & operator[ ](int k); // range-checked indexing

// modifiers
void resize(int newRows, int newCols); // resizes matrix to newRows x newCols
// (can result in losing values)
```

Computer Science Answer Key

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1. A	11. C	21. C	31. D
2. E	12. D	22. D	32. D
3. C	13. A	23. C	33. A
4. C	14. A	24. A	34. A
5. E	15. D	25. C	35. B
6. D	16. B	26. D	36. D
7. C	17. A	27. C	37. C
8. D	18. D	28. D	38. E
9. A	19. A	29. E	39. B
10. B	20. B	30. D	40. B